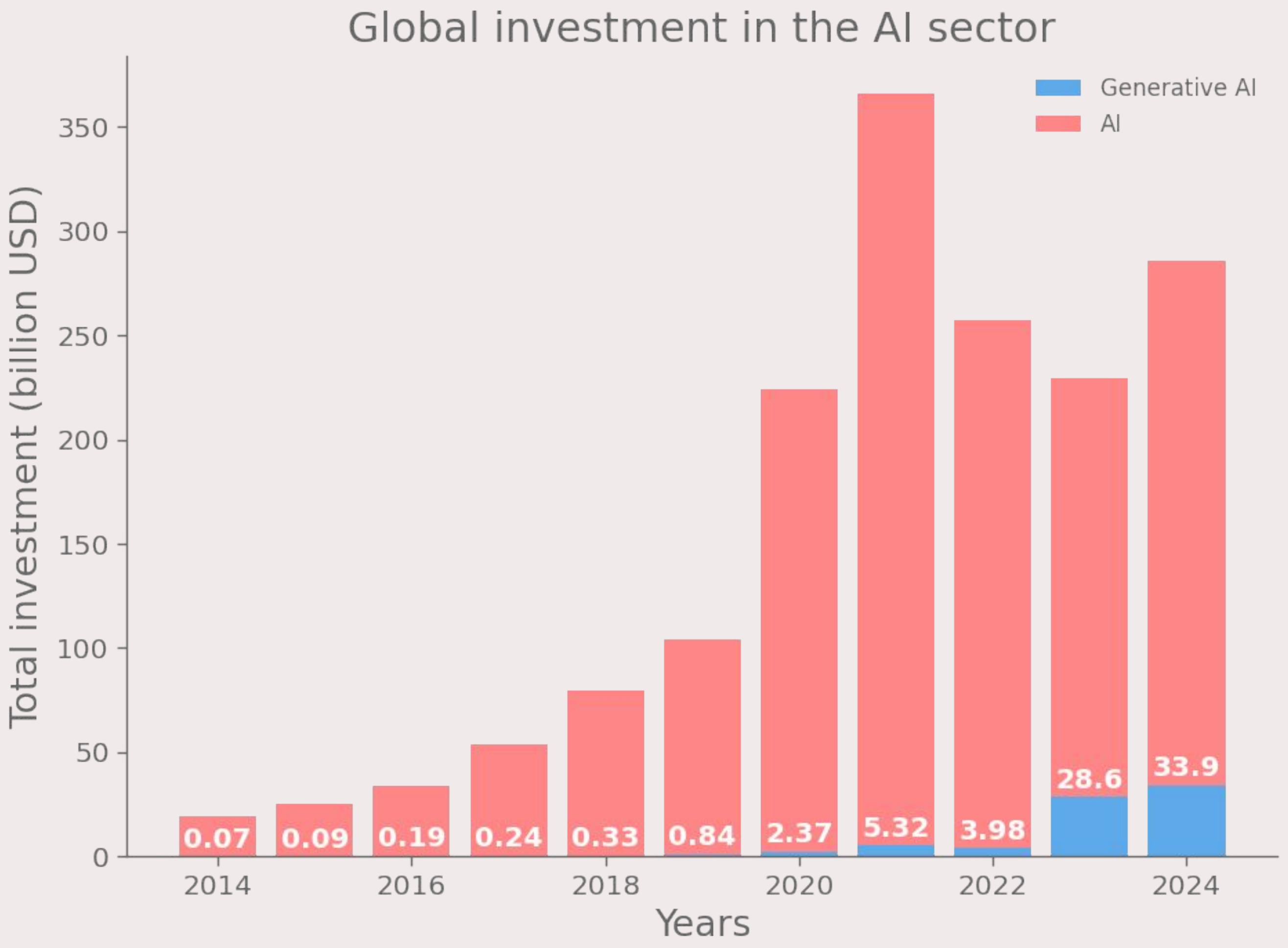
A BRIEF OVERVIEW OF AI

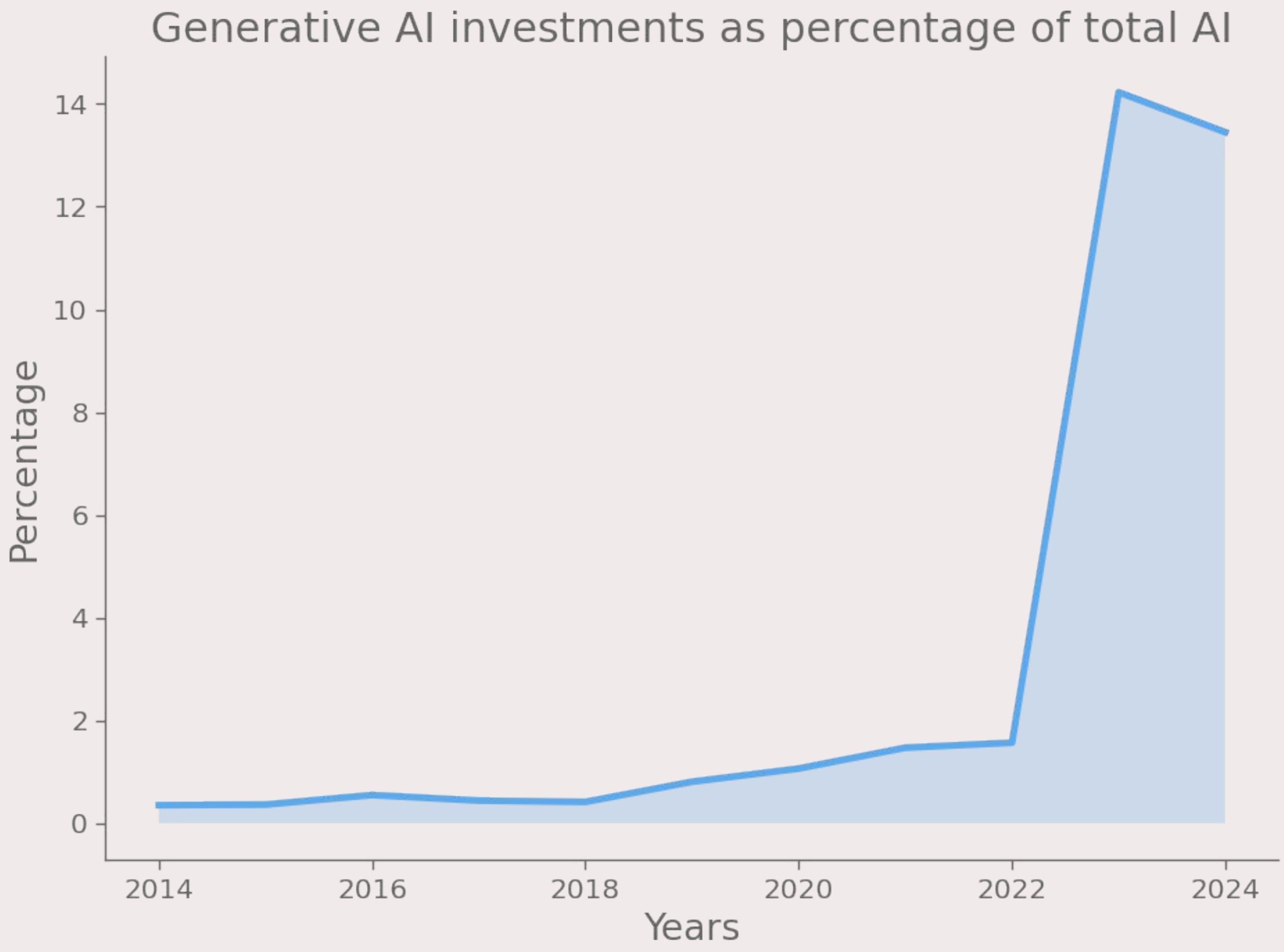
A sector by sector analysis

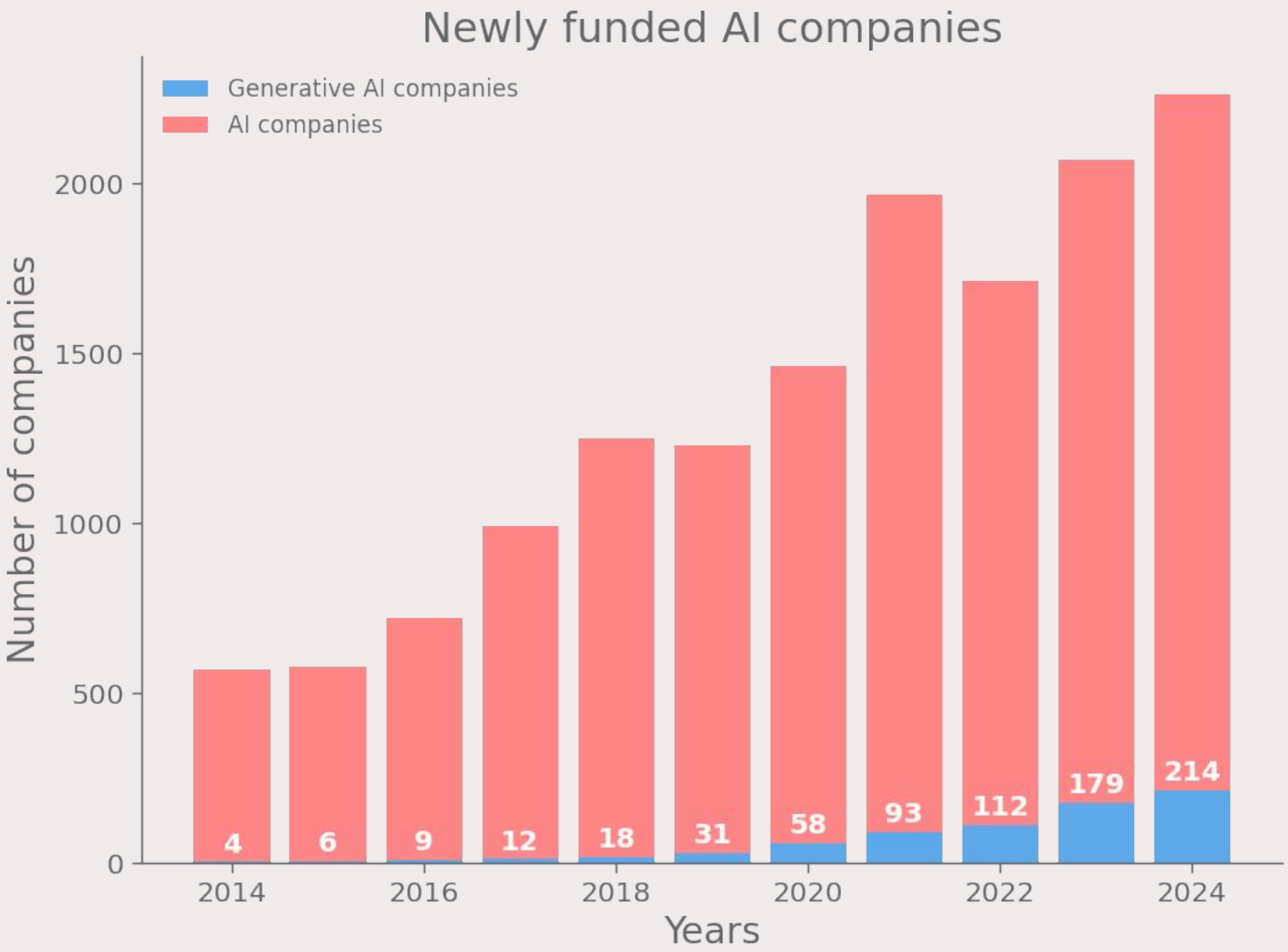
Giovanni Novati 68642A
Simone Alessandro Casciaro 76678A

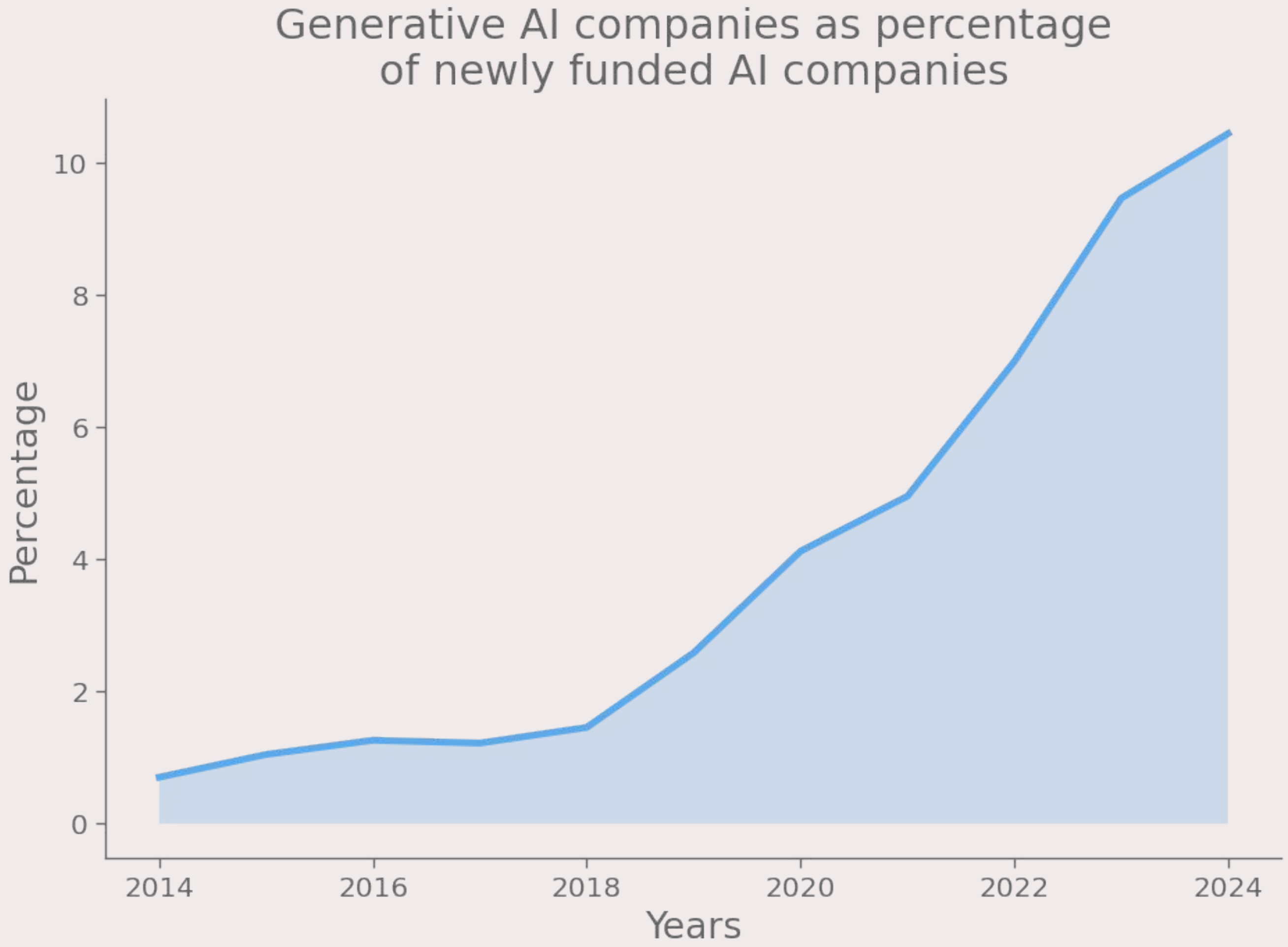
ECONOMY



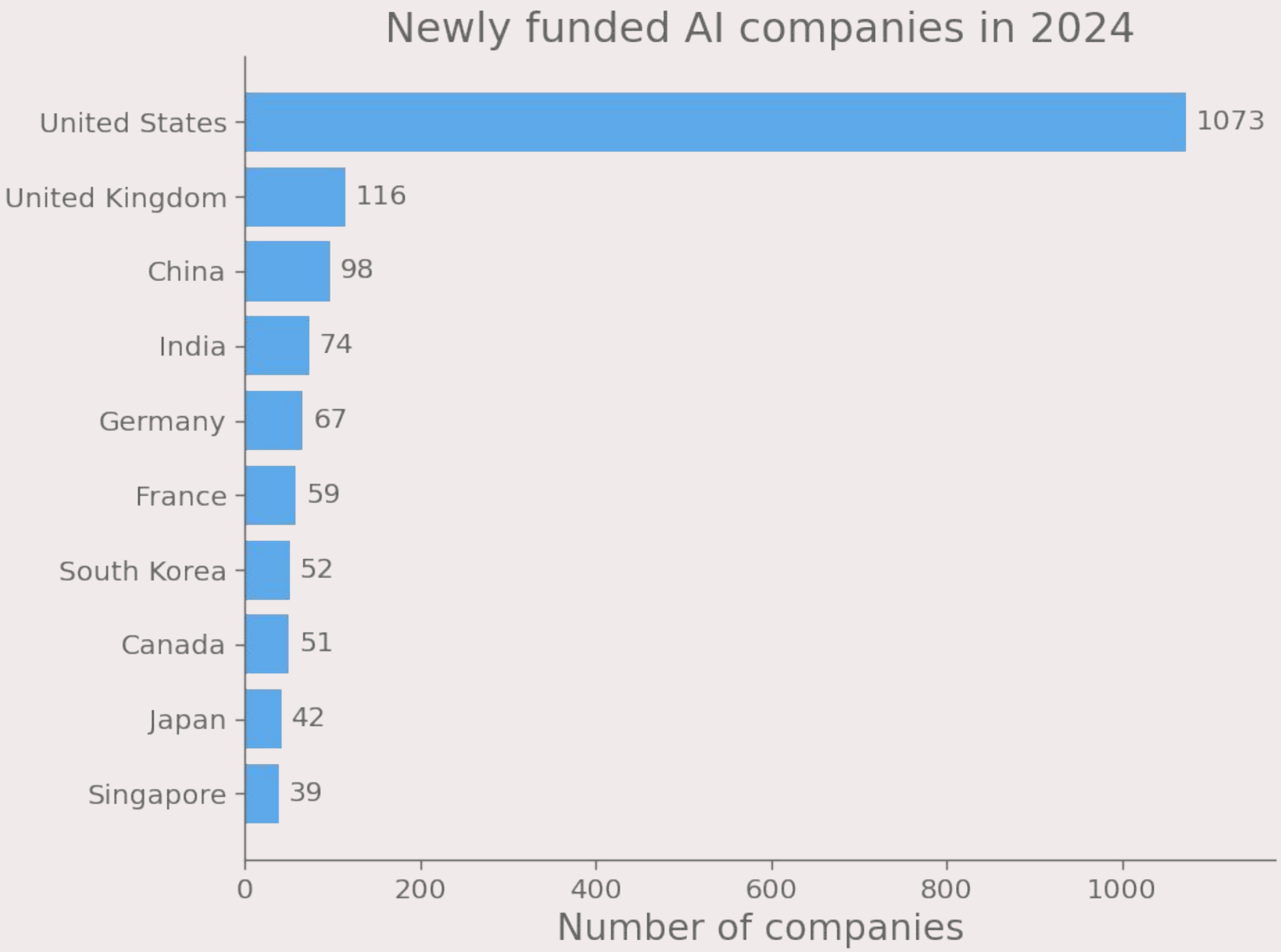
Global investment in the AI sector
Generative AI
AI
Total investment (billion USD)
0
50
100
150
200
250
300
350
0.07
0.09
0.19
0.24
0.33
0.84
2.37
5.32
3.98
28.6
33.9
2014
2016
2018
2020
2022
2024
Years


Generative AI investments as percentage of total AI
Percentage
Years
0
2
4
6
8
10
12
14
2014
2016
2018
2020
2022
2024

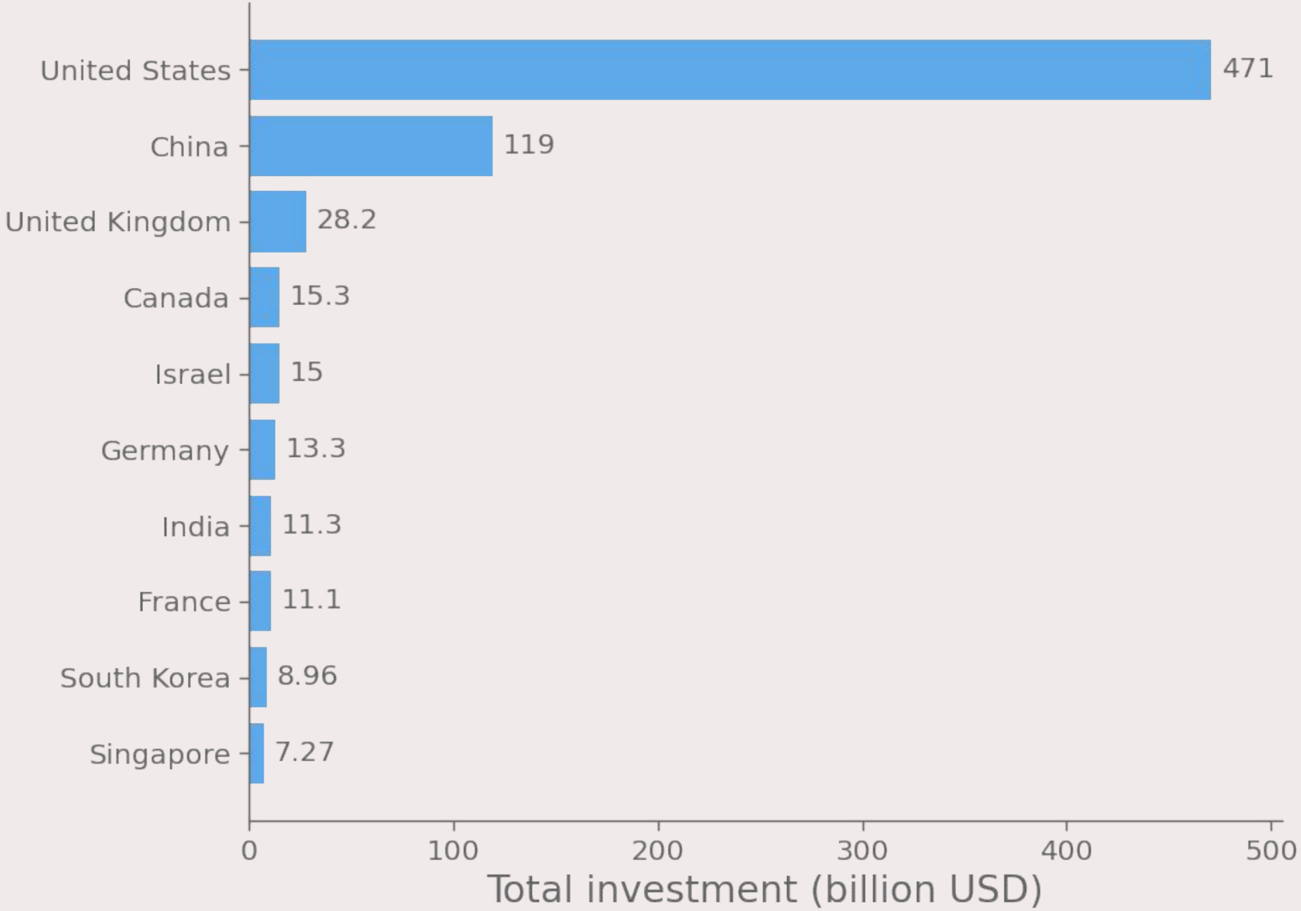

Newly funded AI companies
Generative AI companies
AI companies
Number of companies
2000
1500
1000
500
0
4
6
9
12
18
31
58
93
112
179
214
2014
2016
2018
2020
2022
2024
Years

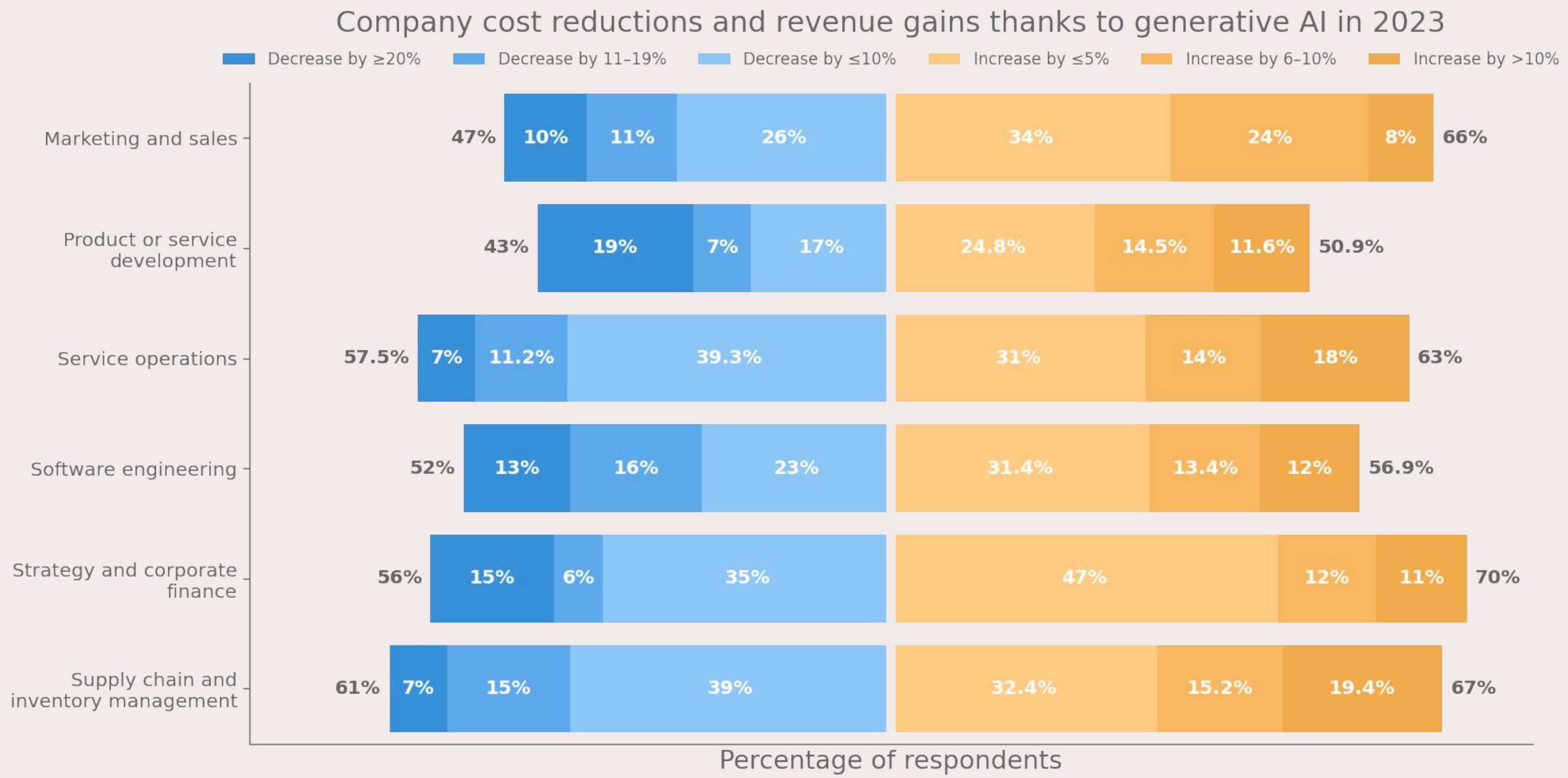


Generative AI companies as percentage
of newly funded AI companies
Percentage
0
2
4
6
8
10
2014
2016
2018
2020
2022
2024
Years


Newly funded AI companies in 2024
United States
United Kingdom
China
India
Germany
France
South Korea
Canada
Japan
Singapore
1073
116
98
74
67
59
52
51
42
39
0
200
400
600
800
1000
Number of companies

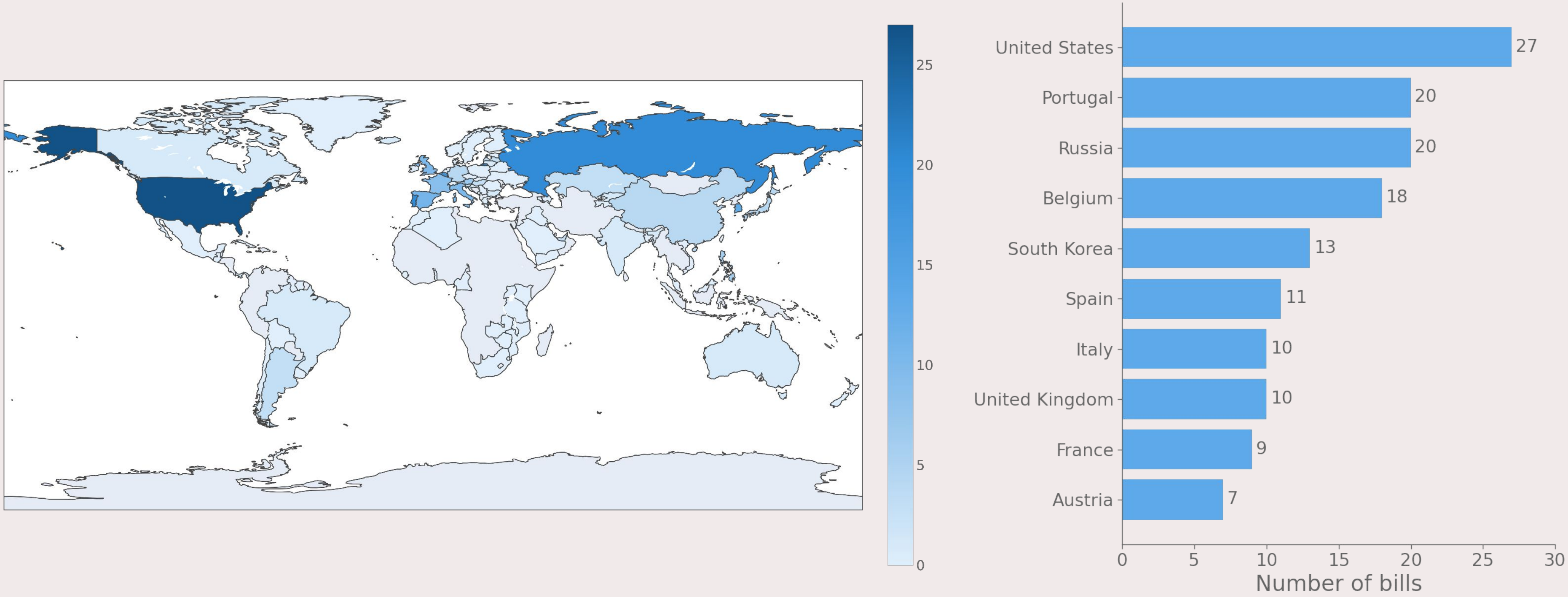
Private investments in AI from 2013 to 2024

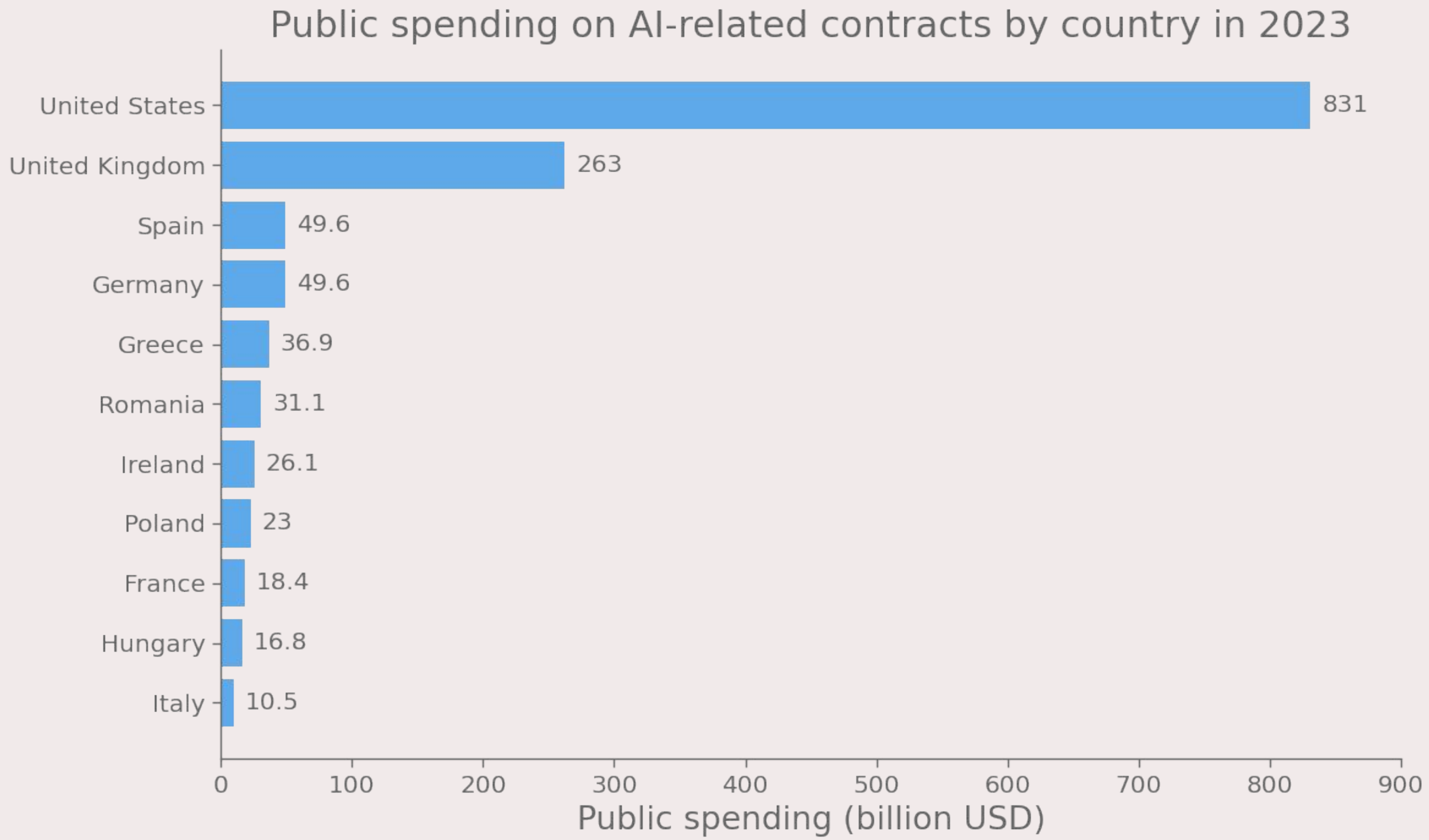



Company cost reductions and revenue gains thanks to generative AI in 2023
Decrease by ≥20%
Decrease by 11–19%
Decrease by ≤10%
Increase by ≤5%
Increase by 6–10%
Increase by >10%
Marketing and sales
47%
10%
11%
26%
34%
24%
8%
66%
Product or service development
43%
19%
7%
17%
24.8%
14.5%
11.6%
50.9%
Service operations
57.5%
7%
11.2%
39.3%
31%
14%
18%
63%
Software engineering
52%
13%
16%
23%
31.4%
13.4%
12%
56.9%
Strategy and corporate finance
56%
15%
6%
35%
47%
12%
11%
70%
Supply chain and inventory management
61%
7%
15%
39%
32.4%
15.2%
19.4%
67%
Percentage of respondents

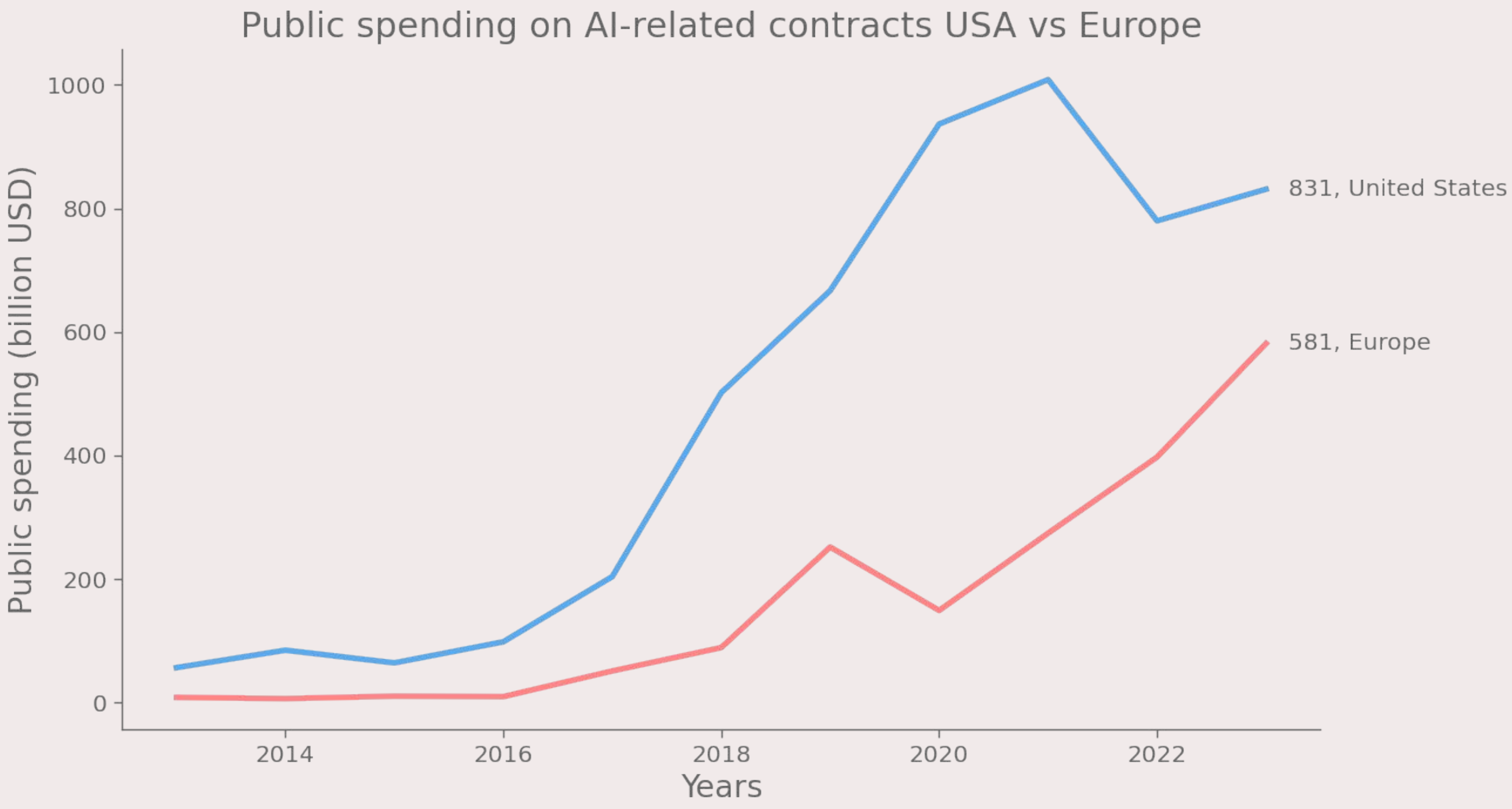
POLICY AND GOVERNANCE

Number of AI-related bills passed into law from 2016 to 2024

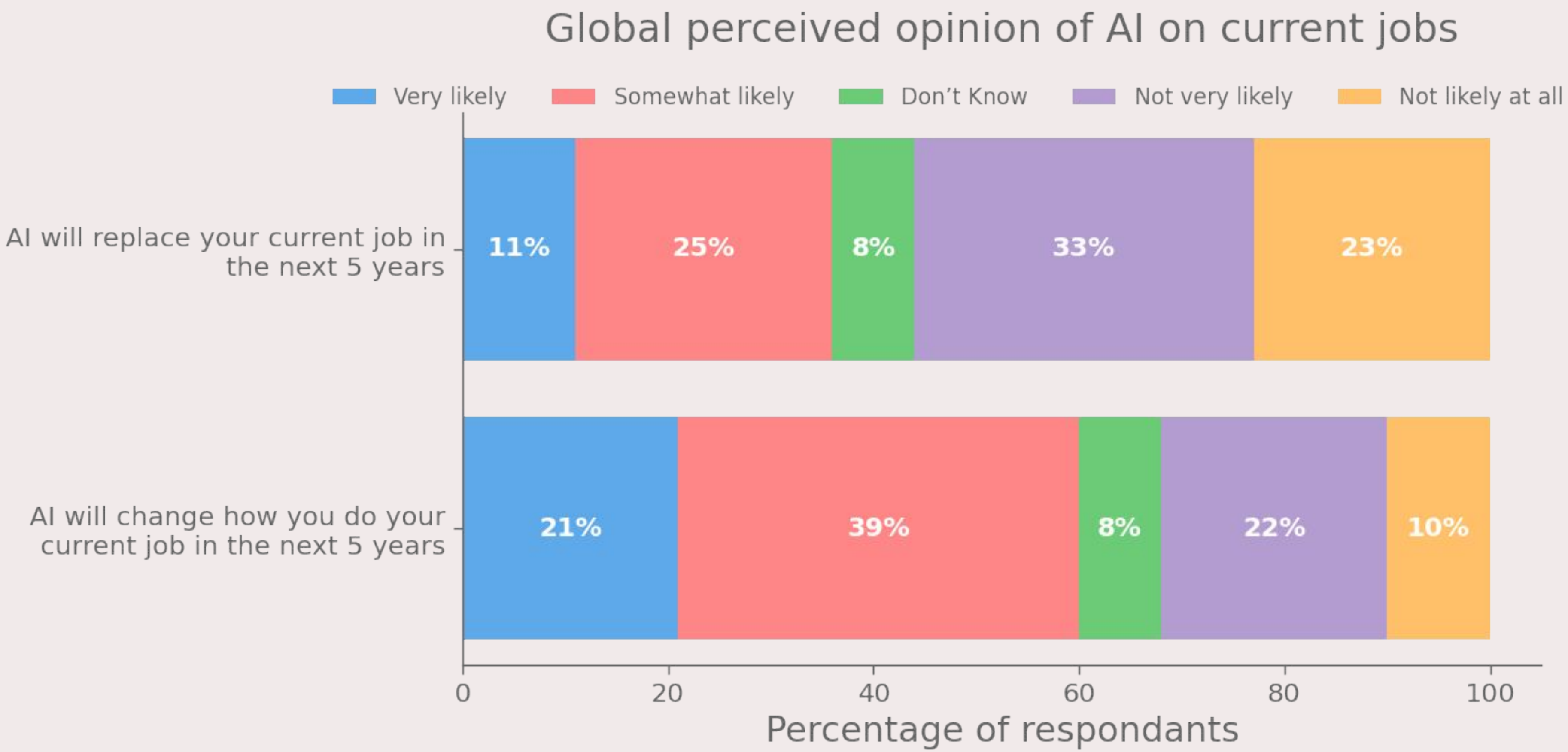


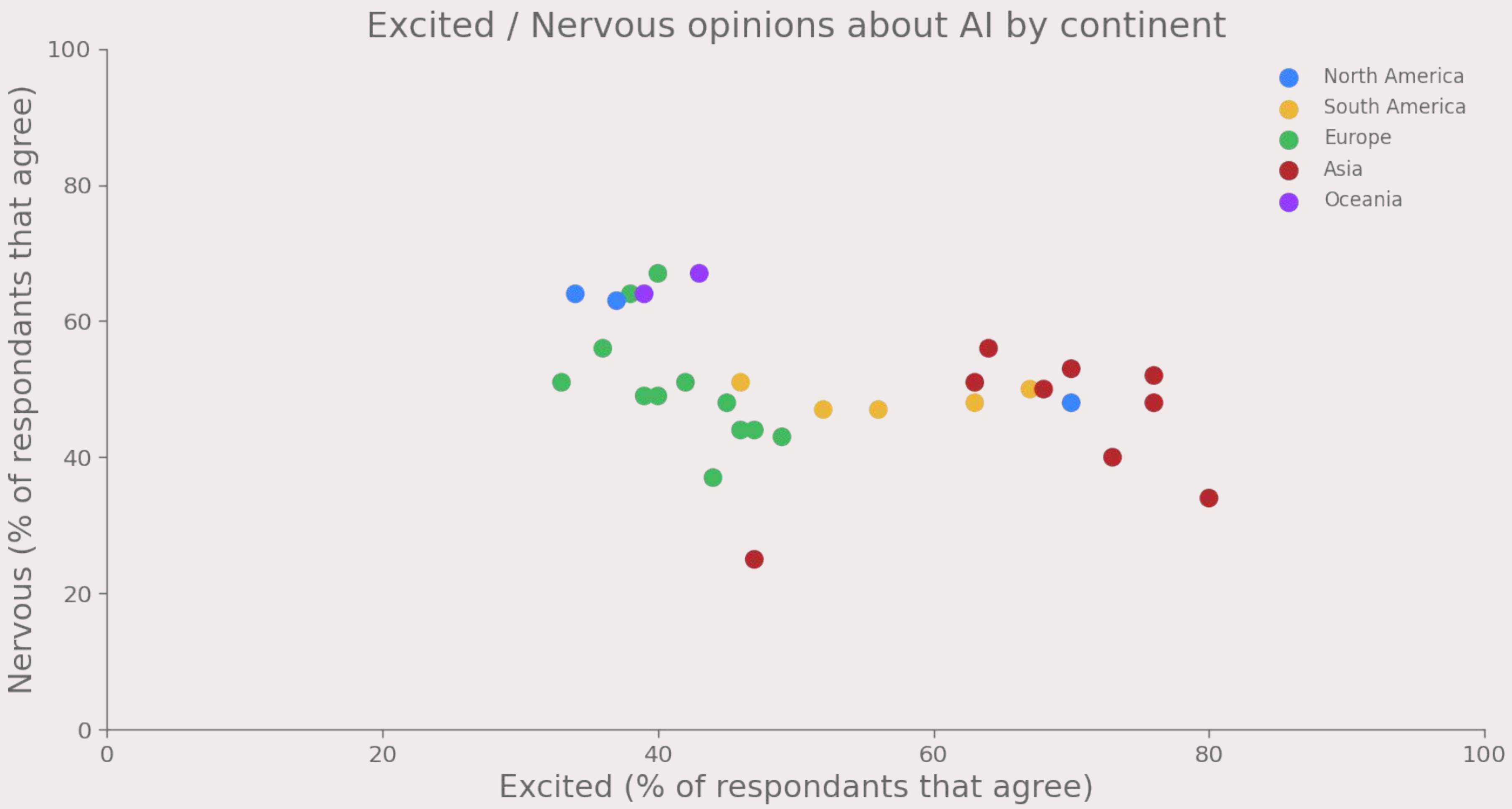


Public spending on AI-related contracts by country in 2023
United States
831
United Kingdom
263
Spain
49.6
Germany
49.6
Greece
36.9
Romania
31.1
Ireland
26.1
Poland
23
France
18.4
Hungary
16.8
Italy
10.5
0
100
200
300
400
500
600
700
800
900
Public spending (billion USD)

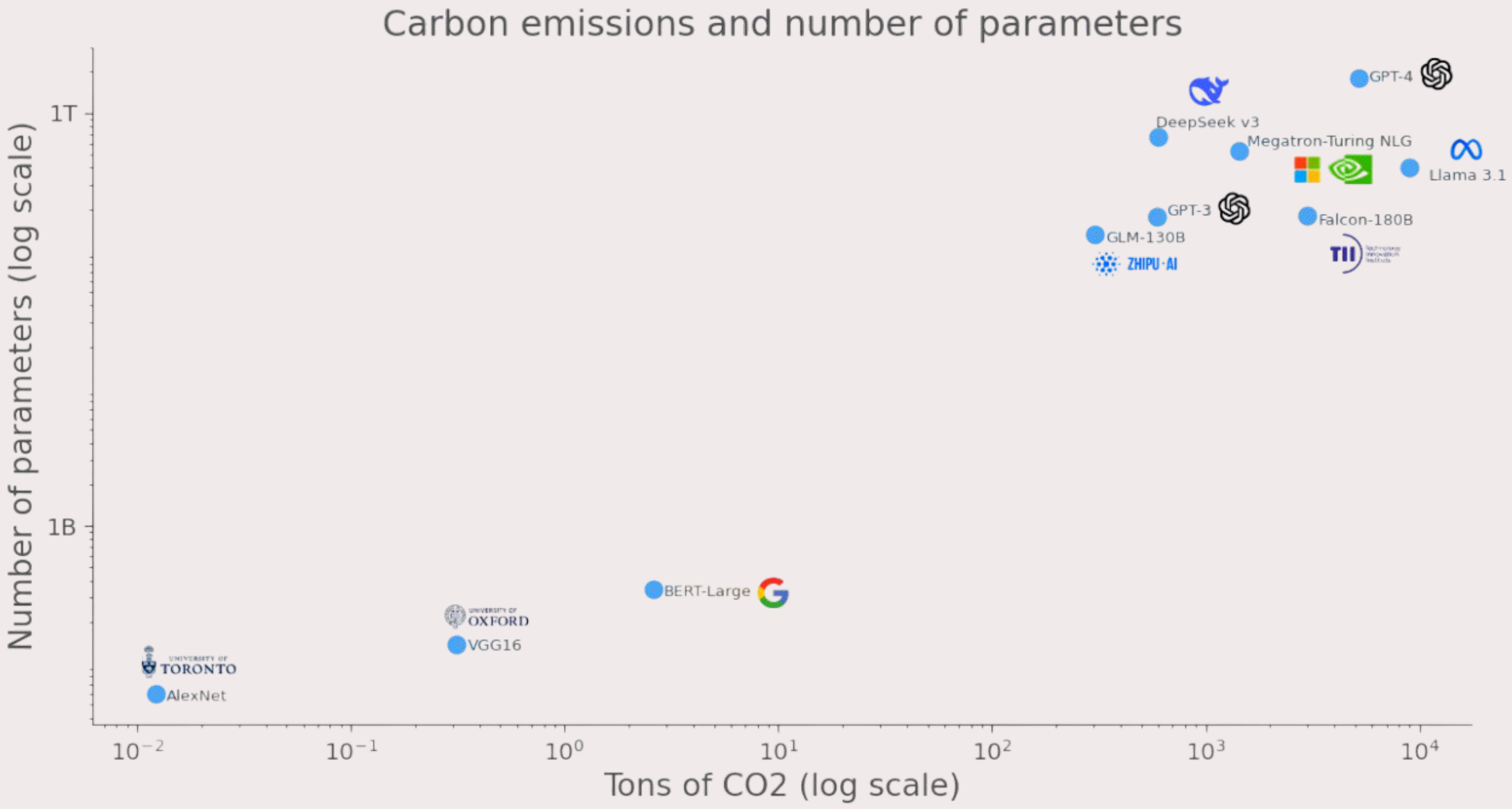


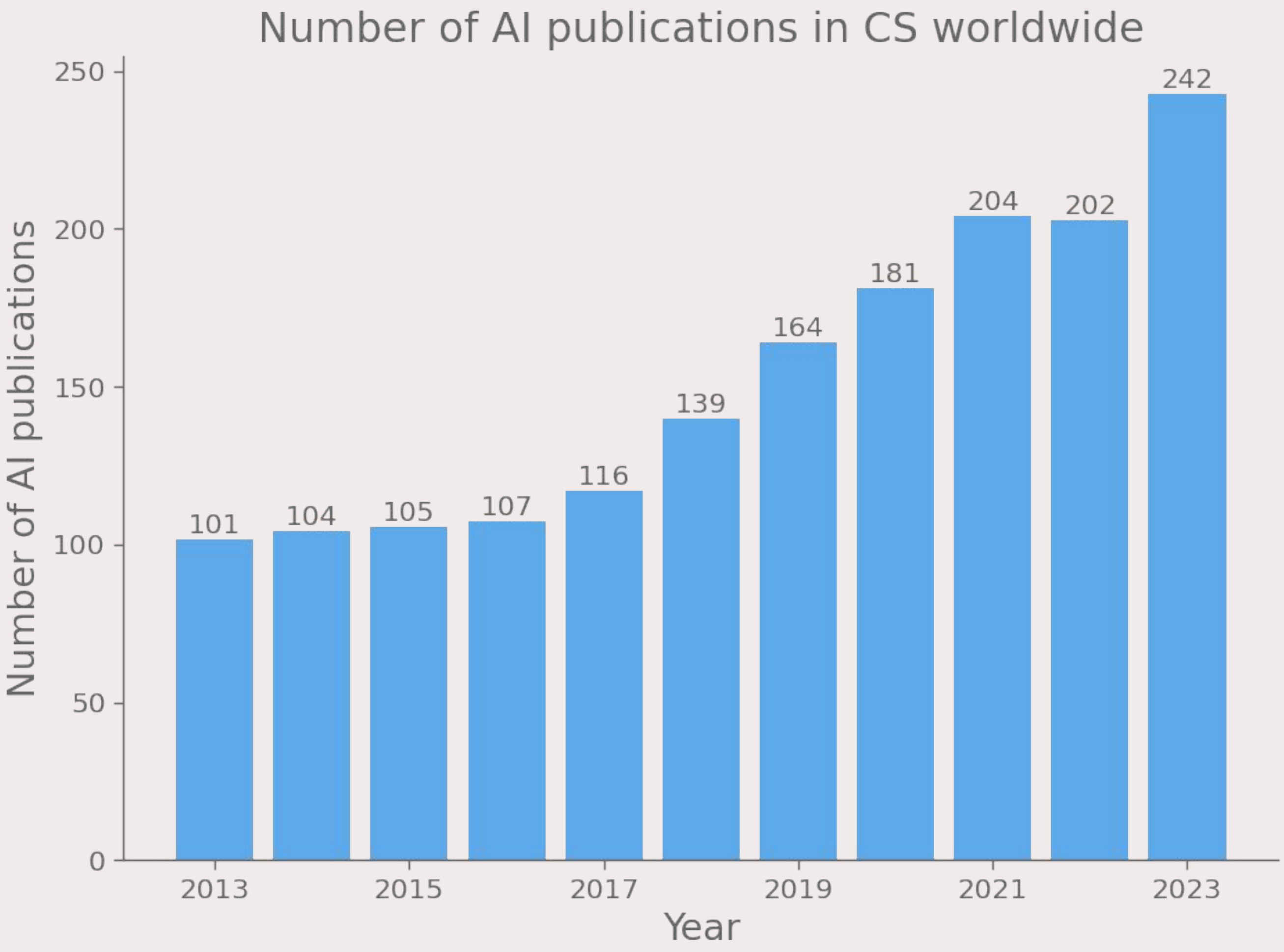
Public spending on AI-related contracts USA vs Europe
Public spending (billion USD)
0
200
400
600
800
1000
2014
2016
2018
2020
2022
Years
831, United States
581, Europe


Global perceived opinion of AI on current jobs
Very likely
Somewhat likely
Don’t Know
Not very likely
Not likely at all
AI will replace your current job in the next 5 years
11%
25%
8%
33%
23%
AI will change how you do your current job in the next 5 years
21%
39%
8%
22%
10%
0
20
40
60
80
100
Percentage of respondants

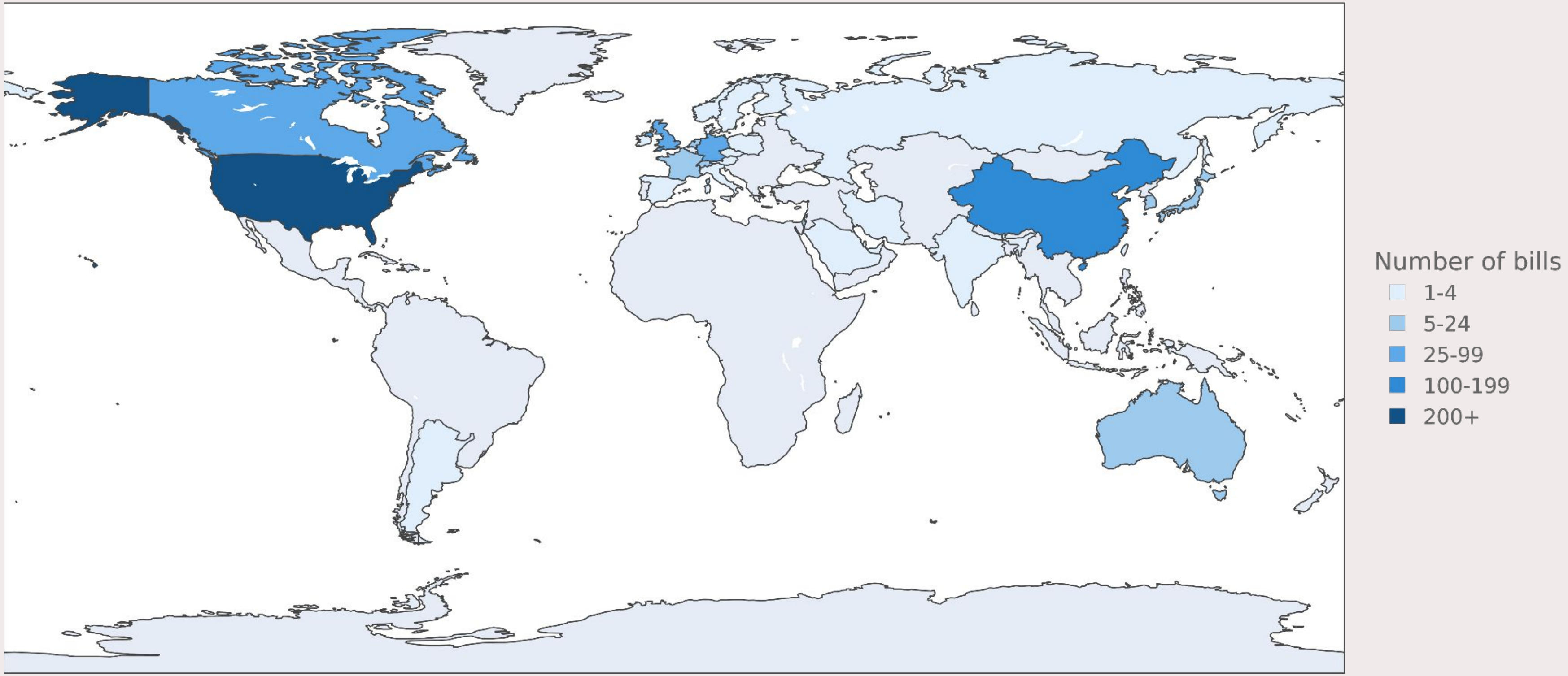

Excited / Nervous opinions about AI by continent
North America
South America
Europe
Asia
Oceania
Nervous (% of respondants that agree)
Excited (% of respondants that agree)
0
20
40
60
80
100

RESEARCH AND DEVELOPMENT

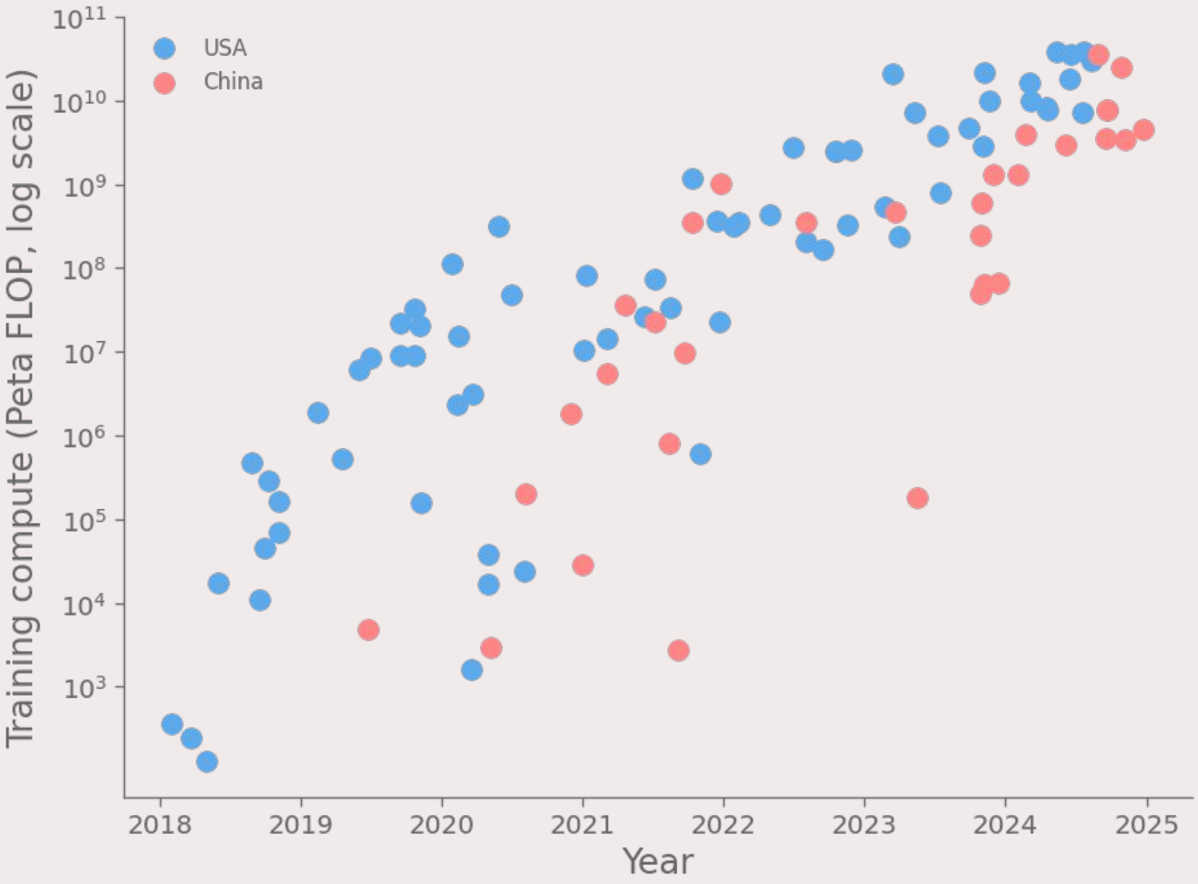

Carbon emissions and number of parameters
Number of parameters (log scale)
1T
1B
Tons of CO2 (log scale)
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
GPT-4
DeepSeek v3
Megatron-Turing NLG
Llama 3.1
GPT-3
Falcon-180B
GLM-130B
ZHIPU·AI
BERT-Large
UNIVERSITY OF OXFORD
VGG16
UNIVERSITY OF TORONTO
AlexNet


Number of AI publications in CS worldwide
Number of AI publications
Year
0
50
100
150
200
250
2013
2015
2017
2019
2021
2023
101
104
105
107
116
139
164
181
204
202
242

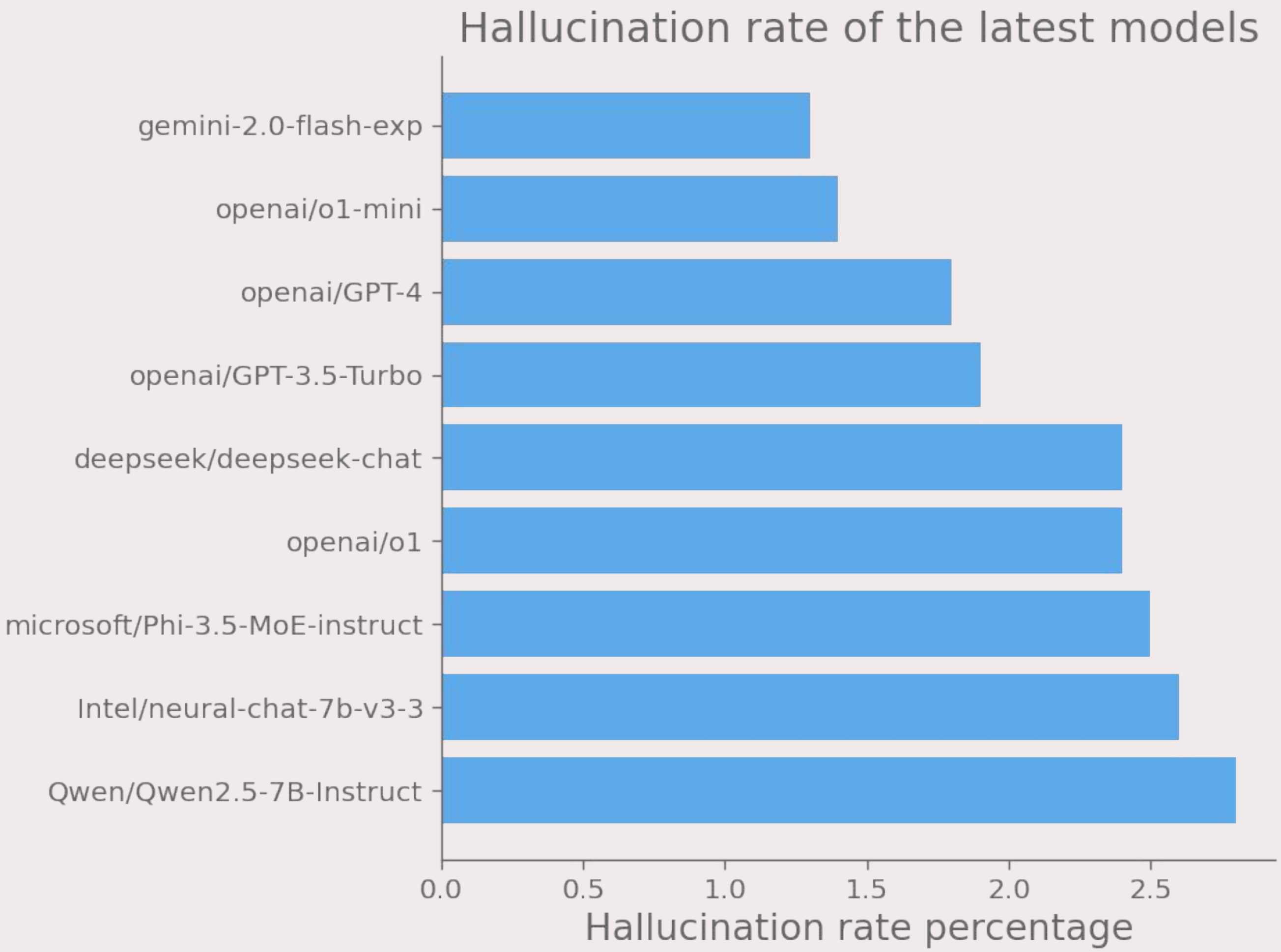
Number of AI-related bills passed into law from 2016 to 2024



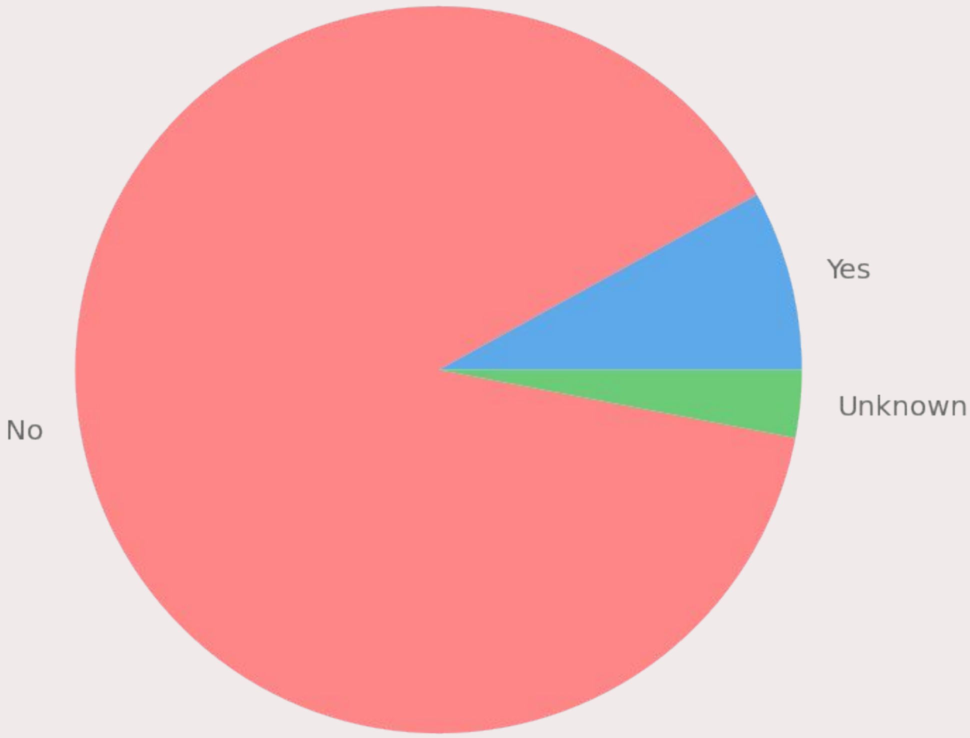
Training compute of select notable AI models in the United States and China

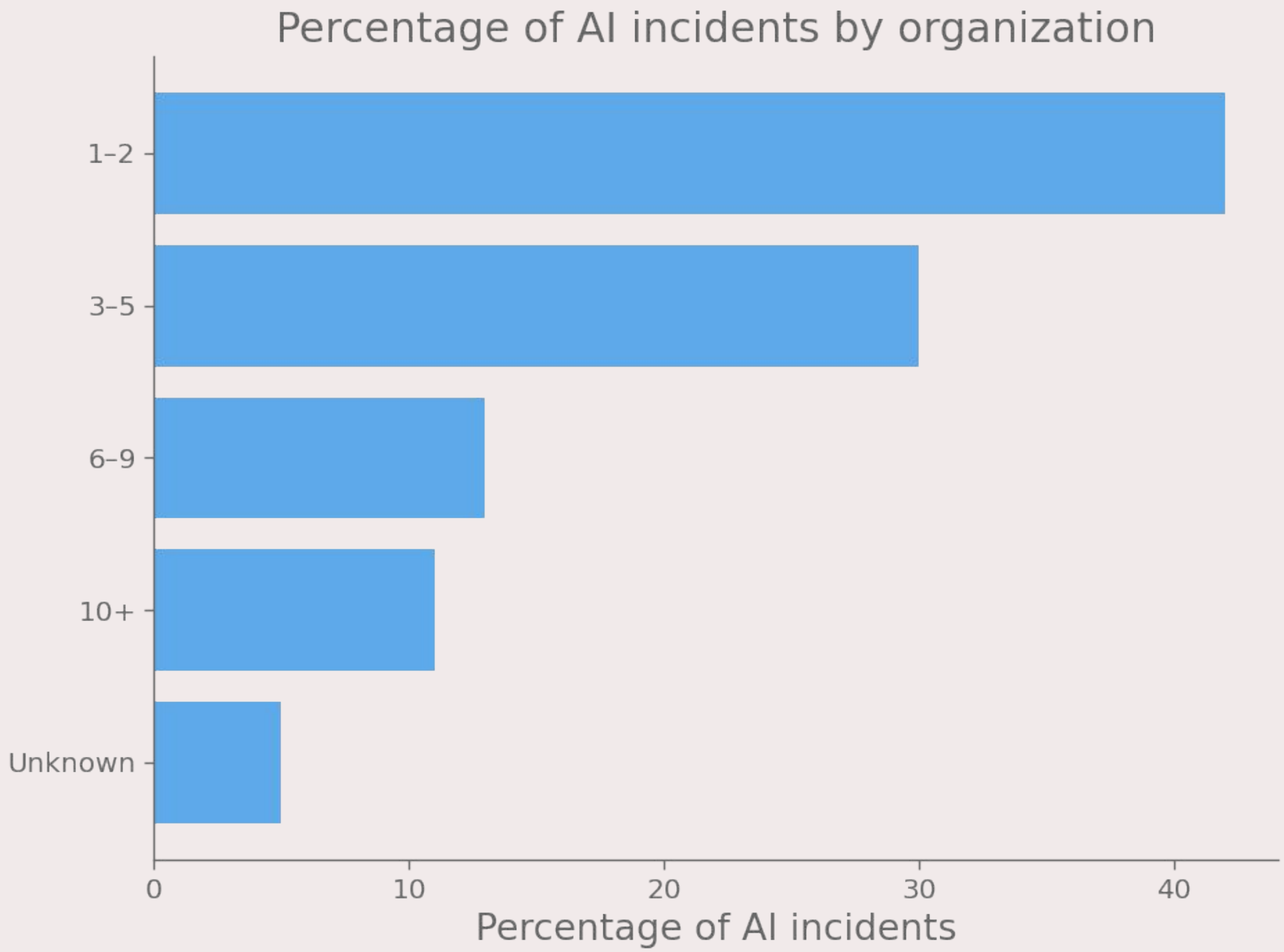


RESPONSIBLE AI

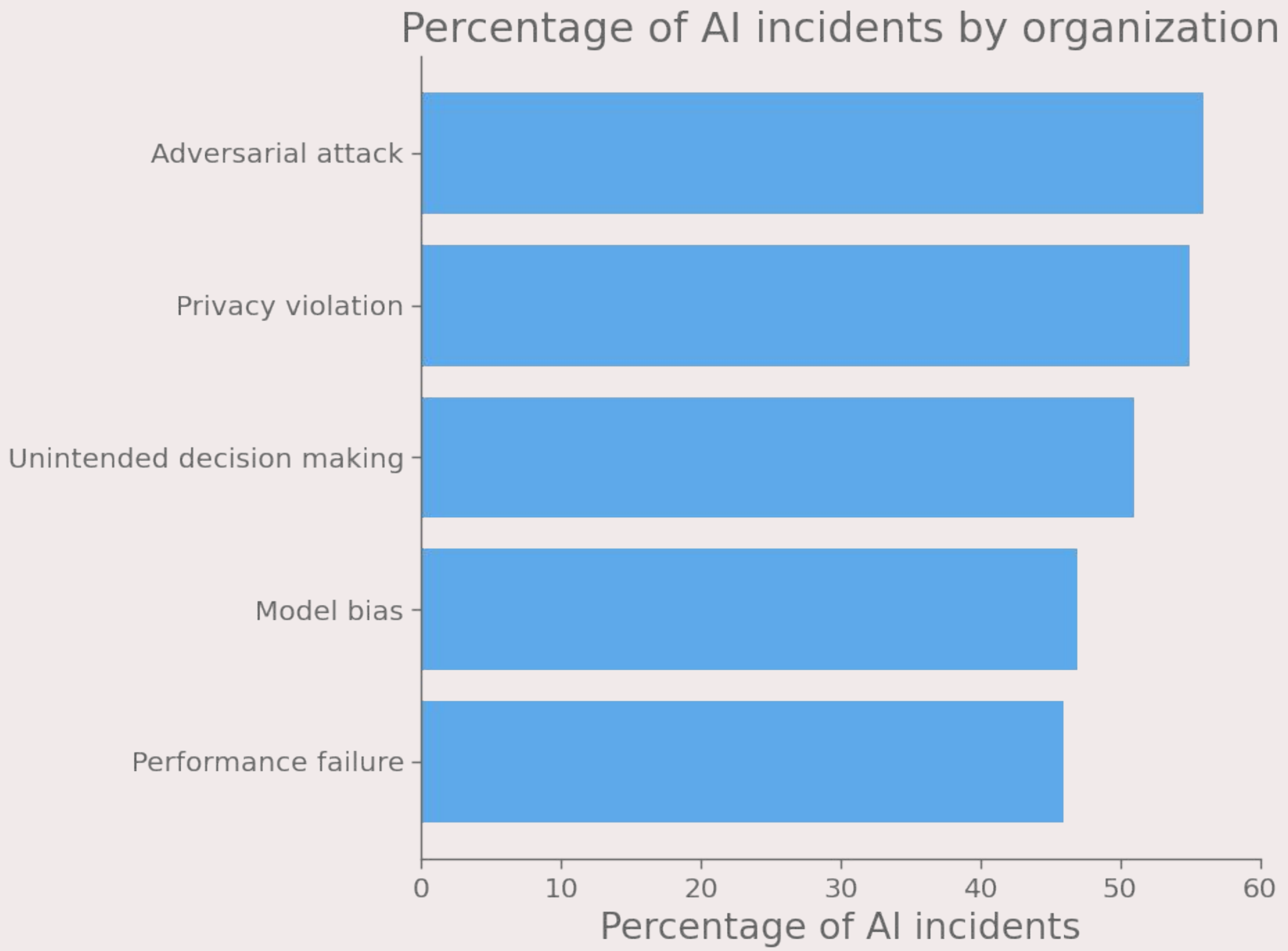

Hallucination rate of the latest models
gemini-2.0-flash-exp
openai/o1-mini
openai/GPT-4
openai/GPT-3.5-Turbo
deepseek/deepseek-chat
openai/o1
microsoft/Phi-3.5-MoE-instruct
Intel/neural-chat-7b-v3-3
Qwen/Qwen2.5-7B-Instruct
0.0
0.5
1.0
1.5
2.0
2.5
Hallucination rate percentage

Percentage of organizations who experienced AI incidents



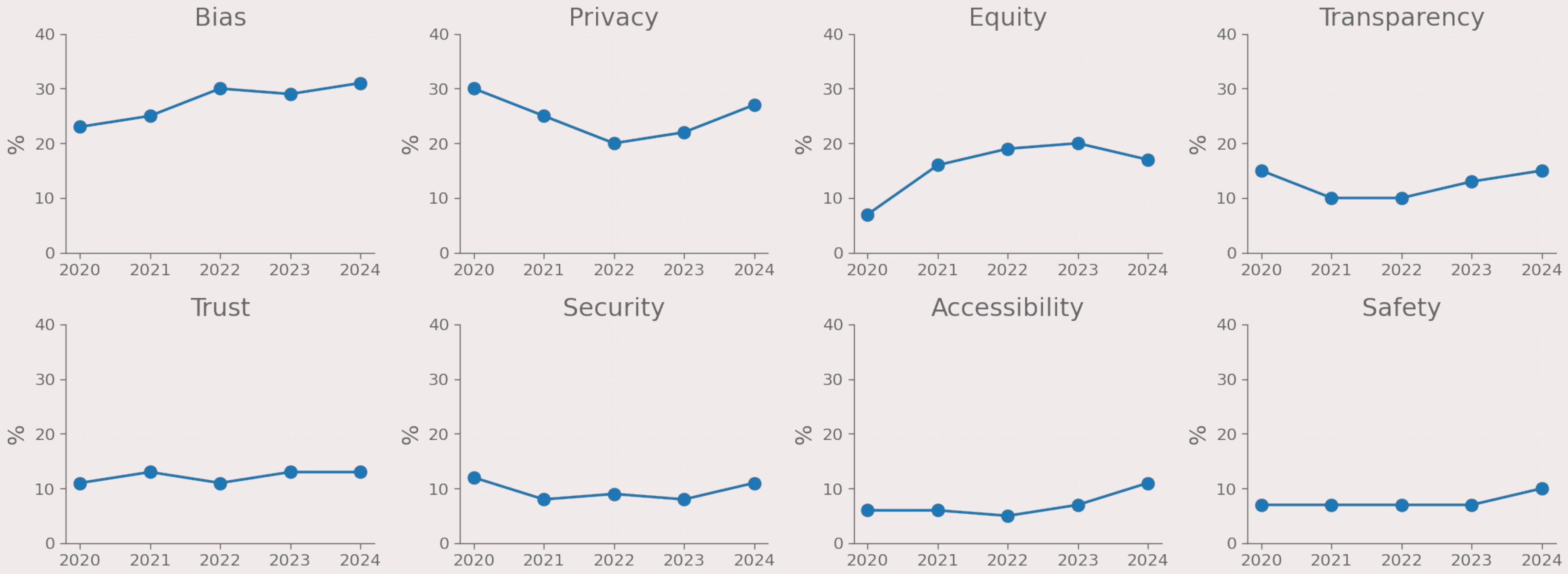


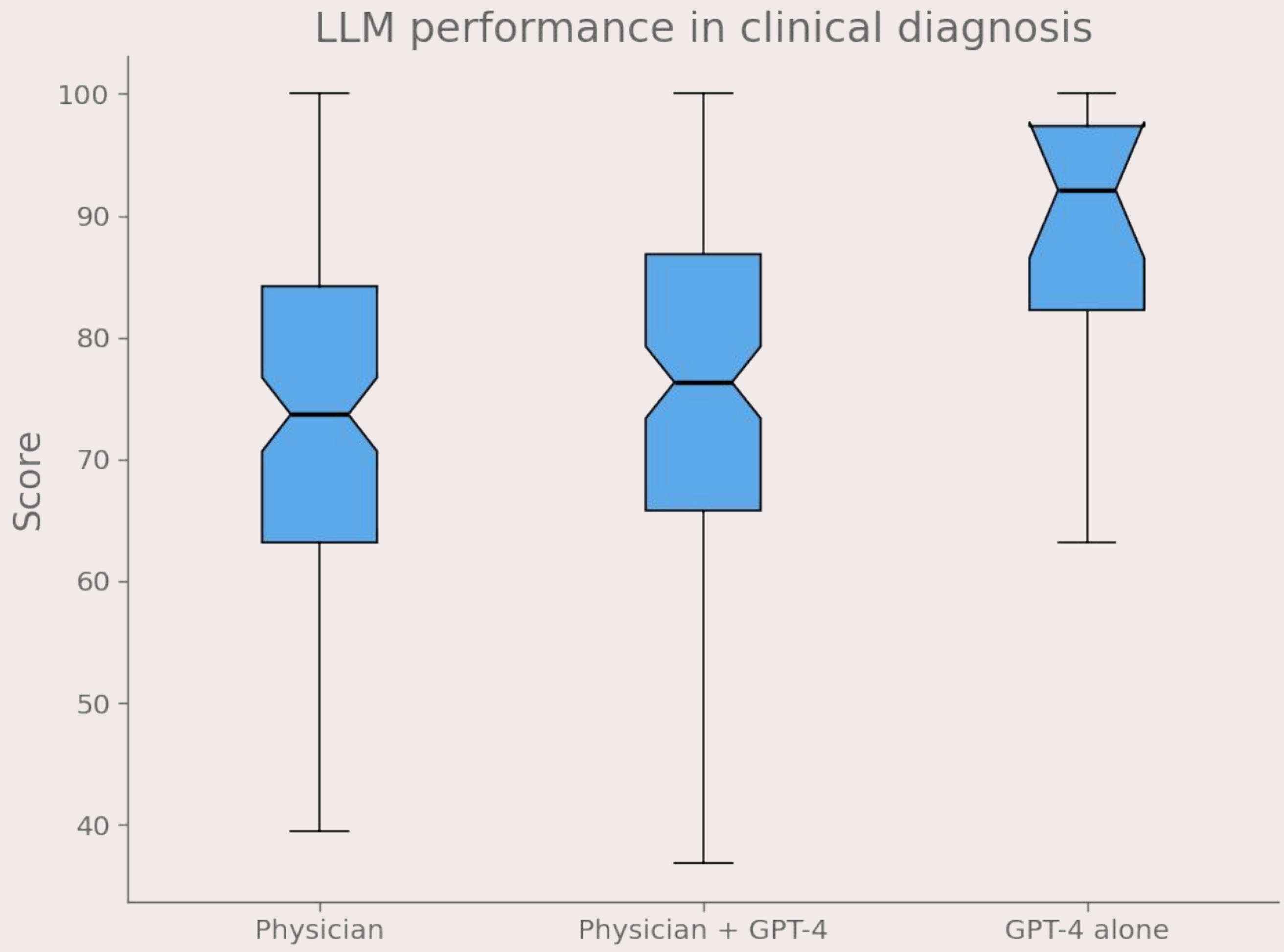
Percentage of AI incidents by organization
1–2
3–5
6–9
10+
Unknown
0
10
20
30
40
Percentage of AI incidents


Percentage of AI incidents by organization
Adversarial attack
Privacy violation
Unintended decision making
Model bias
Performance failure
0
10
20
30
40
50
60
Percentage of AI incidents

SCIENCE AND MEDICINE

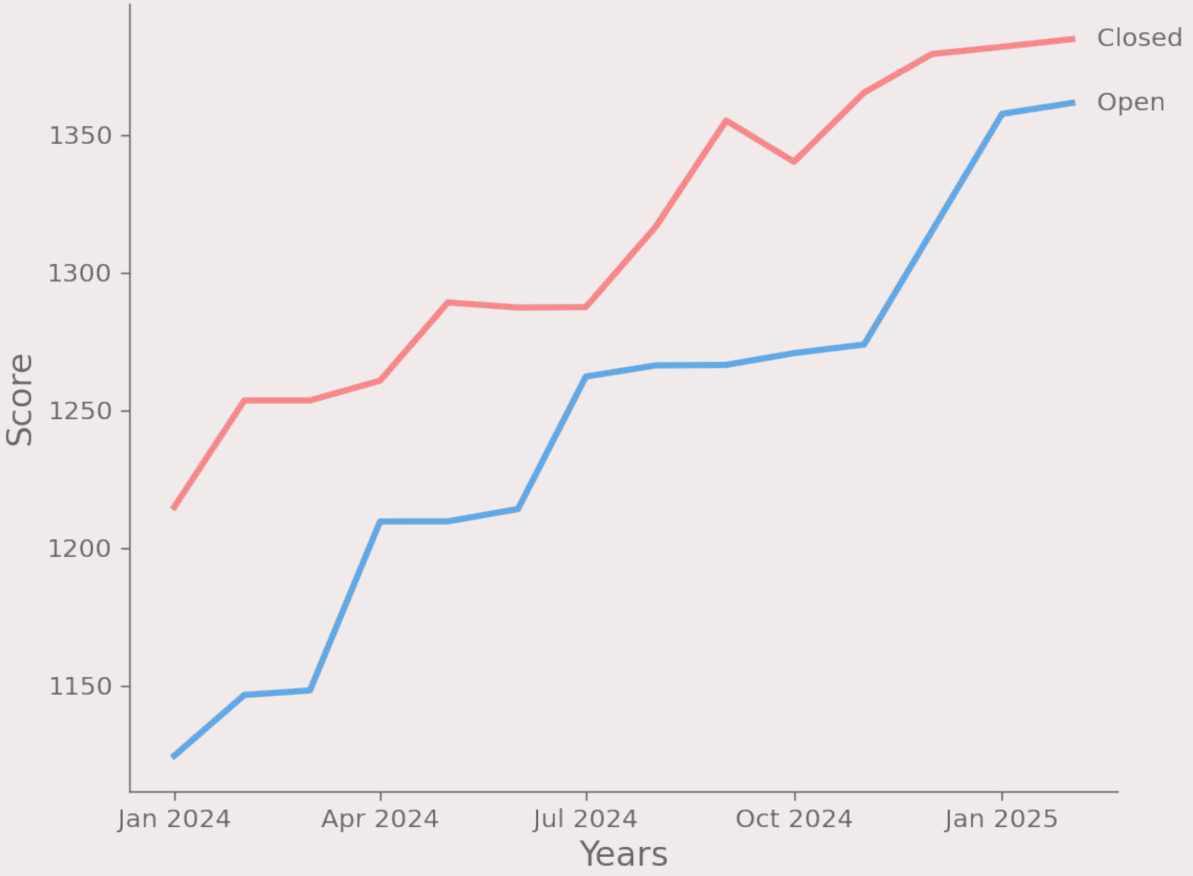
Top 8 ethical concerns discussed in medical AI ethics publications

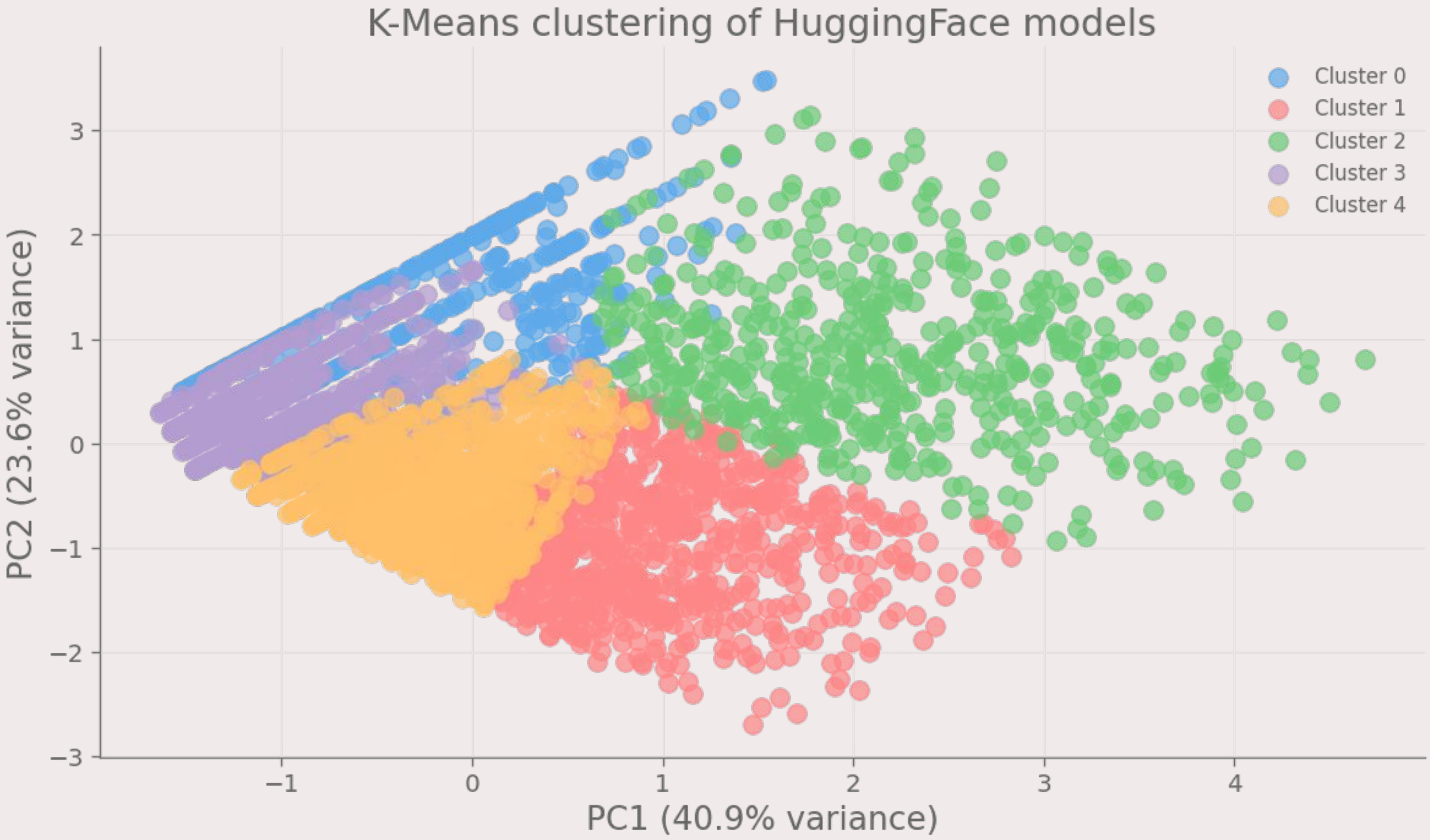



LLM performance in clinical diagnosis
Score
100
90
80
70
60
50
40
Physician
Physician + GPT-4
GPT-4 alone

TECHNICAL PERFORMANCE

Performance of top closed/open models over years




K-Means clustering of HuggingFace models
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
PC2 (23.6% variance)
PC1 (40.9% variance)
3
2
1
0
−1
−2
−3
−1
0
1
2
3
4

Cluster	Size	Avg. Downloads	Avg. Likes	Top Architecture	Top Task
0 (●)	481	6590	2	timm	image classification
1 (●)	863	3012	52	transformers	text generation
2 (●)	555	53766	30	transformers	text generation
3 (●)	1418	1655	1	transformers	text generation
4 (●)	1569	1604	7	transformers	text generation

FONTI

https://hai.stanford.edu/ai-index/2025-ai-index-report
https://www.quid.com/2024-stanford-ai-index-report
https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-2024
https://ifr.org/img/worldrobotics/Press_Conference_2024.pdf
https://www.ipsos.com/en/global-trends-2024
https://www.fda.gov/medical-devices/software-medical-device-samd
https://med.stanford.edu/raisehealth.html
https://epoch.ai/data/machine-learning-hardware
https://www.ipsos.com/en/global-trends-2024
https://zenodo.org/records/14801921

GRAZIE PER L’ATTENZIONE